# TITLE- ONLINE SHOPPERS PURCHASING INTENTION ANALYSIS

**DOMAIN- DATA SCIENCE**

**NAME- HIMANI JESWANI**

**MENTOR- THOMAS ANTONY**

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# INTRODUCTION

- **Online shopping platforms generate large amounts of user data.**
- **Understanding purchasing behavior helps improve sales and user experience.**
- **This project predicts whether a visitor will make a purchase.**
- **Analyze online shopper behavior.**
- **Perform exploratory data analysis (EDA).**
- **Build a machine learning model to predict purchasing intention.**
- **Evaluate model performance.**

# DATASET DESCRIPTION

- **Dataset: Online Shoppers Purchasing Intention**
- **Rows: 12,330**
- **Columns: 18**
- **Target column: Revenue (True = purchase, False = no purchase)**
- **Contains numerical and categorical features.**

# EXPLORATORY DATA ANALYSIS

**Exploratory Data Analysis (EDA) was performed to understand the structure of the dataset.**

- **The dataset contains 12,330 rows and 18 columns.**
- **There are no major missing values, so all columns are usable.**
- **Features include numerical and categorical data describing visitor behavior.**
- **The target variable Revenue has True / False values, indicating purchasing intention.**
- **Statistical summary shows numerical features like Administrative, Informational, ProductRelated, and categorical features like Month and VisitorType.**

# DATA PREPROCESSING

- **Convert categorical variables into numerical format.**
- **Check and handle missing values (if any).**
- **Split dataset into training and testing sets.**

[3]:

```
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder

# Convert Revenue column to 0 and 1
df['Revenue'] = df['Revenue'].astype(int)

# Encode categorical columns
le = LabelEncoder()

df['Month'] = le.fit_transform(df['Month'])
df['VisitorType'] = le.fit_transform(df['VisitorType'])
df['Weekend'] = le.fit_transform(df['Weekend'])

# Define features and target
X = df.drop('Revenue', axis=1)
y = df['Revenue']

# Split data into training and testing
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

print("Preprocessing Done ✅")
```

```
Preprocessing Done ✅
```

# Model Building

- Use a classification model (like Logistic Regression or Decision Tree).
- Train on training dataset.
- Test on unseen data.
- Predict customer purchase intention.

[2]:
```
import pandas as pd

# Load dataset
df = pd.read_csv('/kaggle/input/datasets/imakash3011/online-shoppers-purchasing-intention-dataset/onlin

# Show first 5 rows
df.head()
```

[2]:

| | Administrative | Administrative_Duration | Informational | Informational_Duration | ProductRelated | ProductRelated_Duration | BounceRa |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 0.0 | 0 | 0.0 | 1 | 0.000000 | ( |
| 1 | 0 | 0.0 | 0 | 0.0 | 2 | 64.000000 | ( |
| 2 | 0 | 0.0 | 0 | 0.0 | 1 | 0.000000 | ( |
| 3 | 0 | 0.0 | 0 | 0.0 | 2 | 2.666667 | ( |
| 4 | 0 | 0.0 | 0 | 0.0 | 10 | 627.500000 | ( |

# Results

- **Model achieves good accuracy (mention approximate 85–87% if you want).**
- **Correctly classifies purchasing and non-purchasing visitors.**

[4]:

```python
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report

# Create model
model = LogisticRegression(max_iter=1000)

# Train model
model.fit(X_train, y_train)

# Predictions
y_pred = model.predict(X_test)

# Accuracy
accuracy = accuracy_score(y_test, y_pred)
print("Model Accuracy:", accuracy)

# Detailed report
print("\nClassification Report:\n")
print(classification_report(y_test, y_pred))
```

```
Model Accuracy: 0.8706407137064072
```

# CONCLUSION

- **EDA helped understand visitor behavior.**
- **The machine learning model predicts purchasing intention effectively.**
- **The project can help businesses improve marketing and sales strategies.**
- **Future improvements: use advanced models and feature engineering.**